

1 Rollout

policy

π_t



transitions

GAE

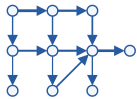
A_r, A_{c_k}

episode cost

$\bar{J}_k$

2 Flow inputs

state clusters



transition edges



$\hat{F}$
flow



A
reward



C_k
costs

3 Set constraints

Soft constraints

$$b_k = B_k/L$$

$$\langle C_k, F \rangle \leq b_k$$



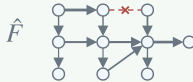
Hard constraints



$$F_H = 0$$

4 Entropic projection

One flow solve



target flow
is not executed

$$\min_{F \geq 0} -\langle A, F \rangle$$

$$+\varepsilon \text{KL}(F \| \hat{F})$$

s.t. flow conservation

5 Priced policy update

Safety prices

$$\lambda_k^* \rightarrow \bar{\lambda}_k^*$$

clip at $c + \text{EMA}$

observed-cost feedback

$$\beta_k \leftarrow [\beta_k + \alpha(\bar{J}_k - B_k)]_+$$

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO/TRPO $\rightarrow \pi_{t+1}$

observed episode costs

GAE advantages

next rollout